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Rethinking Offshore Power Smart EHV Solutions for a Greener Future

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Abstract

EHV networks at offshore fields require long Submarine Cables, Gas Insulated Switchgears, EHV Power Transformers, Shunt Reactors and associated Control & Protection Systems. Due to the dielectric properties of SF6 gas, EHV Busducts are traditionally used for interconnection of EHV Power Equipment. This paper highlights methodology followed to minimize the use of EHV Busducts and associated carbon footprint, during recent offshore projects and strategies adopted to mitigate the design concerns. A methodical approach covering all the aspects of the design and installation is discussed to avoid the constraints during the execution stage.

This paper explores the replacement of GIBs with EHV cables on offshore platforms, targeting reduced carbon emissions and improved system efficiency. Subsea Cable routing on the topside of an offshore project is a multi-disciplinary activity that requires close coordination of Electrical, Structural, Piping and Offshore Installation teams. In addition to the permanent items, there are a lot of temporary equipment required to facilitate safe cable pulling offshore. To avoid clashes, cable pulling exclusion volumes need to be considered in 3D Model. Multiple design reviews are undertaken to reach a design acceptable to all the stakeholders. This is extremely challenging without the use of technologies such as Digital Twin with Smart Modeling Techniques.

The study assesses the feasibility of adopting SF6-free EHV solution through Technology-enabled 3D modeling and digital twin simulations. These tools optimize space utilization and predict constructability challenges under offshore constraints, especially during cable pulling through the J Tubes and further routing to GIS in an offshore environment where clashes with equipment and structure are major concerns. Detailed evaluations were conducted to ensure compatibility among GIS, transformers, and reactors, while incorporating smart design principles for operability and maintainability. Stakeholder collaboration was vital in resolving termination complexity and defining robust cable support systems. AI-assisted clash detection and smart configuration mapping improved decision-making, supporting compliance with layout, structural, and safety standards.

Integrating EHV cables impacts engineering and installation phases, especially under deck height constraints due to Minimum Bending Radius requirements and space required for cable pulling equipment / volumes. Smart design simulations identified a 12-meter deck to deck height, optimal for accommodating terminations. A hybrid solution with L-Shape GIB with horizontal EHV cable interface, proved effective,

reducing offshore scope and enhancing layout flexibility. Terminations at transformers/reactors required cable boxes and standard alignment per IEC 62271-209/EN 50299-2, emphasizing the role of precision design and cross-discipline coordination. AI-aided predictive clash analysis helped structural and piping teams adjust their design early in the design phase. High-strength, MBR-compliant supports and trefoil clamp specifications were selected based on data-based installation modeling. Testing posed challenges due to IEC 62067 voltage protocols.

This paper introduces a smart approach to replace GIBs with EHV cables, achieving measurable reductions in carbon footprint and lifecycle costs (CAPEX/OPEX). It highlights the importance of intelligent design tools in ensuring code compliance, optimizing layout, and minimizing operational disruptions. With smart tools, stakeholder collaboration, the design enables easier maintenance and better integration across disciplines. The resulting system supports long-term reliability, efficiency, and environmental goals in offshore electrical infrastructure.

Introduction

The offshore oil and gas industry is undergoing a paradigm shift towards sustainable energy practices, driven by increasing environmental regulations, corporate social responsibility, and the need to reduce operational costs while ensuring reliable power delivery in challenging offshore environments. The traditional reliance on GIBs filled with SF₆ poses significant environmental and logistical challenges due to SF₆'s high Global Warming Potential and the complexities in manufacturing, installation, and maintenance. The integration of EHV cable systems, augmented by advanced digital engineering tools including digital twin simulations and smart modelling, offers a promising alternative. This paper presents an in-depth study of replacing GIBs with EHV cables in a 230 kV offshore network, highlighting the comprehensive design, engineering, installation, and testing methodologies that ensure operational reliability while reducing environmental impact.

Offshore platforms are complex systems requiring reliable high-voltage power distribution for drilling, processing, accommodation, and safety-critical systems. Electrical infrastructure must address:

- Space and weight restrictions due to platform design
- Harsh environmental conditions including salt spray, humidity, and vibrations
- Stringent safety standards mandated by international standards
- Maintenance constraints imposed by offshore accessibility and weather windows

Traditionally, GIS and GIBs have been favored for their compactness and insulation properties.

- **SF₆ Gas and Environmental Implications** SF₆, while offering excellent dielectric and arc-quenching properties, is one of the most potent greenhouse gases. Its GWP is approximately 23,500 times that of CO₂ over 100 years. Even small leakages during the lifecycle—from manufacture, installation, operation, to decommissioning—can lead to significant emissions. The European Union and other jurisdictions have enacted strict regulations limiting SF₆ use and mandating leak detection and repair, thus increasing operational overhead.
- **Limitations of GIB Systems in Offshore Context** GIBs require meticulous assembly and alignment, making offshore installation highly specialized and time-consuming. They necessitate continuous gas pressure and humidity monitoring, raising maintenance costs and risk of operational downtime. Inflexible layouts restrict design adaptability, which is increasingly important given the dynamic nature of offshore platform expansions or modifications.

Cable Construction and Electrical Properties

EHV cables for 230 kV applications typically consist of:

- Conductor: High conductivity copper or aluminum often compacted and annealed to minimize losses.
- Insulation: Cross-linked polyethylene (XLPE) or ethylene propylene rubber (EPR), designed for high dielectric strength and thermal stability.
- Metallic Screen: Provides electromagnetic shielding and a return path for fault currents.
- Armor and Sheath: Protect against mechanical damage and environmental corrosion, especially vital in offshore installations.

Advantages Over GIBs in General

- Environmental: Zero SF₆ emissions, lower embodied carbon in materials and installation.
- Mechanical Flexibility: Easier accommodation of platform movements and structural deflections.
- Installation Efficiency: Pre-fabrication, simpler terminations, and less offshore labor.
- Maintenance: Reduced inspection frequency and no gas handling.
- Operational Reliability: Lower failure rates due to robust insulation and monitoring.

Smart Design and Engineering Methodology

Cable Routing on an Offshore Platform – A Multi-Disciplinary and Technologically Intensive Process.

Cable routing on an offshore platform is a complex and highly collaborative engineering activity that demands seamless coordination among multiple disciplines, including Electrical, Structural, Piping, and Installation teams. Unlike conventional onshore facilities, the confined and congested layout of offshore platforms imposes additional challenges related to space optimization, safety, and constructability.

This process not only involves the integration of all permanent components—such as cable trays, junction boxes, and electrical equipment—but also necessitates the inclusion of temporary structures that are essential for executing safe and efficient cable pulling operations at sea. These temporary aids may include scaffolding, pulling frames (quadrants), winches, rollers, and rigging points, all of which occupy valuable physical space during installation.

To prevent spatial conflicts and ensure a clash-free design, it becomes crucial to define cable pulling exclusion volumes, zones reserved in the 3D model where no permanent or temporary obstructions are allowed during cable installation. These zones simulate the space required for tension control, bending radii, and pulling clearances, helping to anticipate and mitigate potential issues during the execution phase.

Achieving a design that satisfies all engineering and construction stakeholders requires conducting multiple iterative design reviews. Each review must incorporate feedback from all concerned disciplines, as cable routing can affect or be affected by structural supports, piping runs, HVAC ducts, access routes, and fire safety zones. The iterative nature of this process ensures constructability and safety, but it also demands substantial coordination, effort and decision-making.

Without the use of advanced digital tools and 3D modeling technologies, it is extremely difficult to accurately visualize, simulate, and validate these complex interdependencies. Traditional 2D drawings and disconnected models often lead to design inconsistencies, late-stage clashes, rework, and higher installation risks.

Therefore, embracing intelligent 3D design environments, clash detection software, and digital collaboration platforms become essential for delivering a safe, efficient, and cost-effective cable routing solution on offshore installations. Concept of Digital Twin is briefly defined below.

Digital Twin with Artificial Intelligence Integration. A digital twin interfaced with AI/ ML model is developed with IoT integrated with SCADA/ DCS for building effective simulation model for different stress and failure analysis using Edge Computing, based on Vendor Data. An integrated 3D Model connected with the EPC Digital Twin is then developed after clash resolution. An Electrical Cable Design system (Aveva Electrical/SPEL) is used to integrate with 3D model to route the cable with optimization rules. This has helped in establishing a clear correlation with Cable Trays and Support supporting the Path of Construction. With this optimization, an effective Drum Plan can be developed integrated with RPA based Supply chain simulation for effective material delivery for constraint free construction supporting better predictability and productivity with increased safety in Construction. 4D/5D solutions integrated with Digital Twin are used for Kanban based reporting. Augmented and Virtual Reality integrated with Digital Twin and Predictive analysis is used during execution and review, reducing manning in the field and supporting reduced Carbon Emission in travelling. Virtual Reality is effective in analyzing the different risk scenarios that may arise during commissioning and startup.

Once the plant Operation will start and the data historian will have collected data at least for a year analyze performance, the predictive model created will be used for generating and triggering notifications to CMMS for maintenance.

A digital twin of the entire electrical system is developed incorporating:

- 3D models of structural, electrical, and mechanical components
- Real-time data during commissioning and operation
- Simulation modules for mechanical and electrical analysis

This twin can be updated iteratively with vendor data, enabling:

- Early clash detection and spatial conflict resolution
- Simulation of cable routing and installation processes
- Predictive maintenance modeling based on operational data

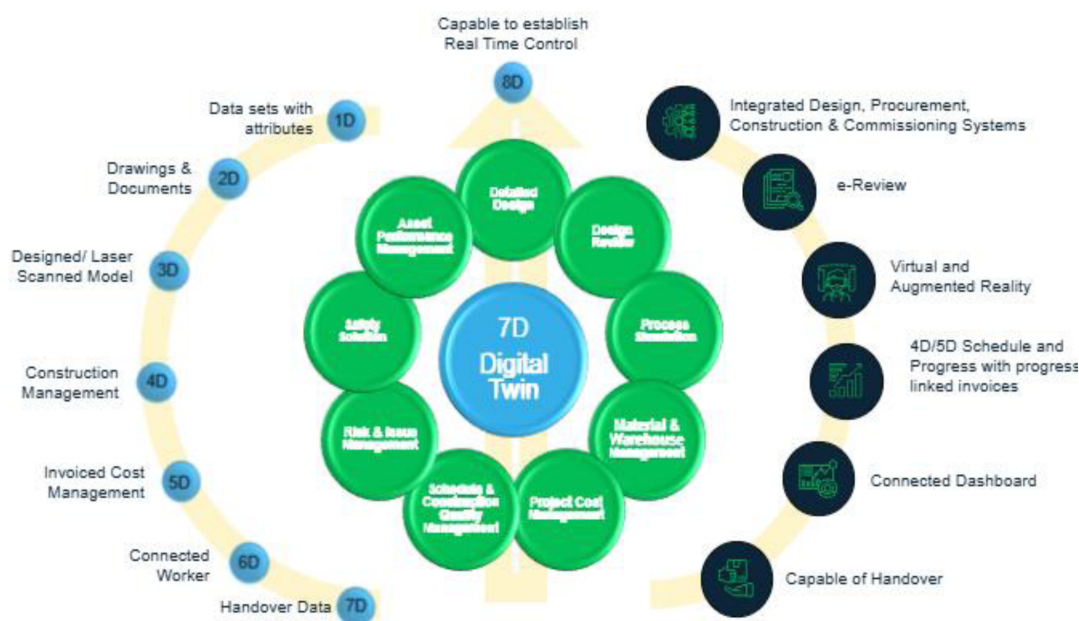


Figure 1—Schematic of the digital twin environment

Clash Detection and Compliance Checking. Integrated software platforms provided automated clash detection against the following:

- Structural elements
- HVAC ducting and piping
- Topsides Equipment
- Safety and maintenance access zones

Project Details

NMDC Energy executed an oil and gas project wherein 230kV power supply is fed from onshore substation to an offshore field through 3Cx630 sq.mm. submarine composite cable with 2x50 FO cores. Project involved a 230kV submarine cable ring network with seven segments feeding four Oil Platforms and one Power Platform. Each platform has slightly different configuration, however they can be classified into two types. Platform-2 is one type with GIS and Power Transformers / Shunt Reactors located on different deck levels, while these are located on the same deck level at Platform-1/3/4/6. Platform 2 is the existing platform that was installed by us in the past based on the GIB concept. Refer Figure 2 below for the 230kV Submarine Cable Ring Network.

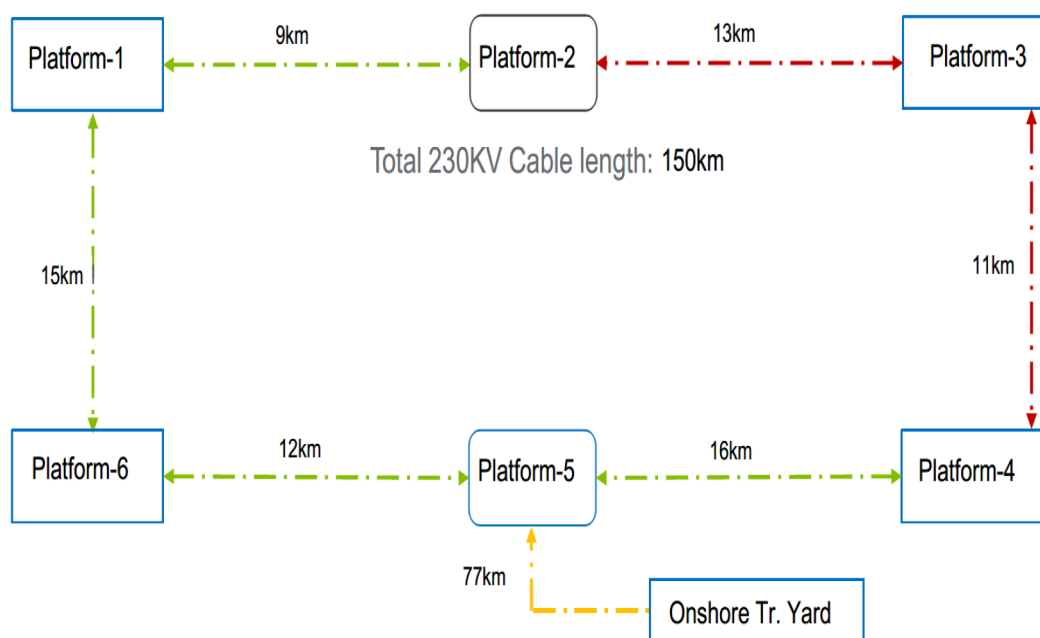


Figure 2—230kV Submarine Cable Ring Network

The overall diameter of the submarine cable is 229.7mm with a weight of 89.1kg in air and 53.9kg in water. Refer Figure 3 for submarine composite cable details.

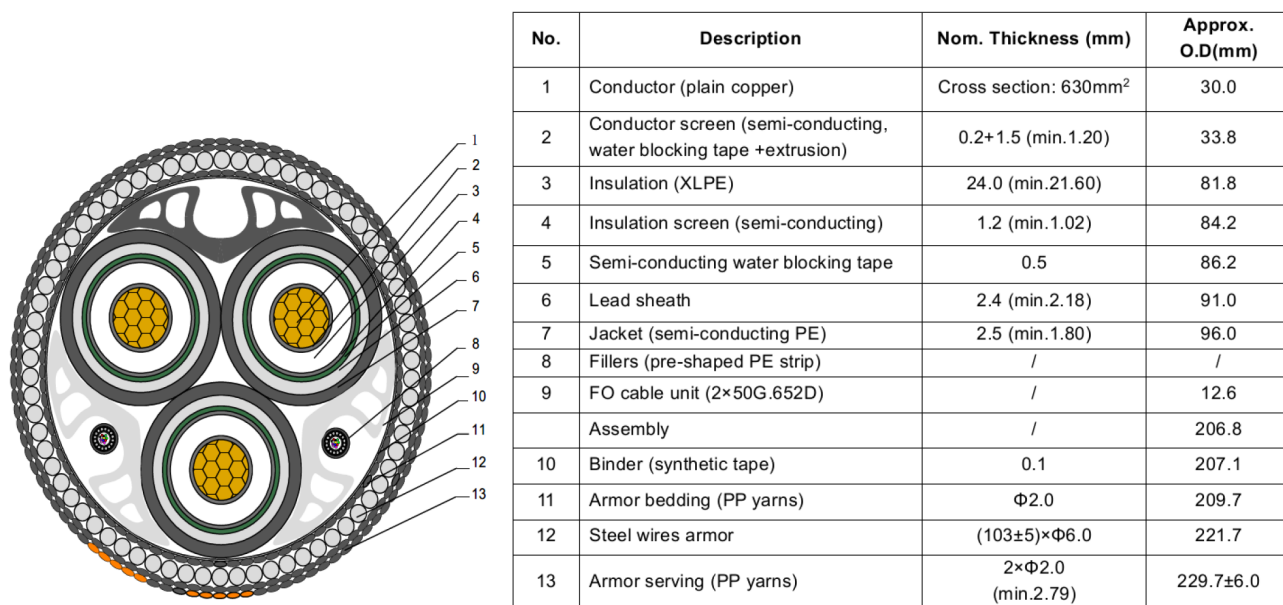


Figure 3—Submarine Composite Cable Cross-Section

Conventional Design

Considering that handling voltages at EHV levels requires use of high dielectric mediums, GIBs have been traditionally utilized as part of electrical power distribution systems. Figure 4 below indicates the conventional design with GIB arrangement that was considered as part of the FEED based on which Contractor had to perform Detailed Engineering, Procurement and Construction activities.

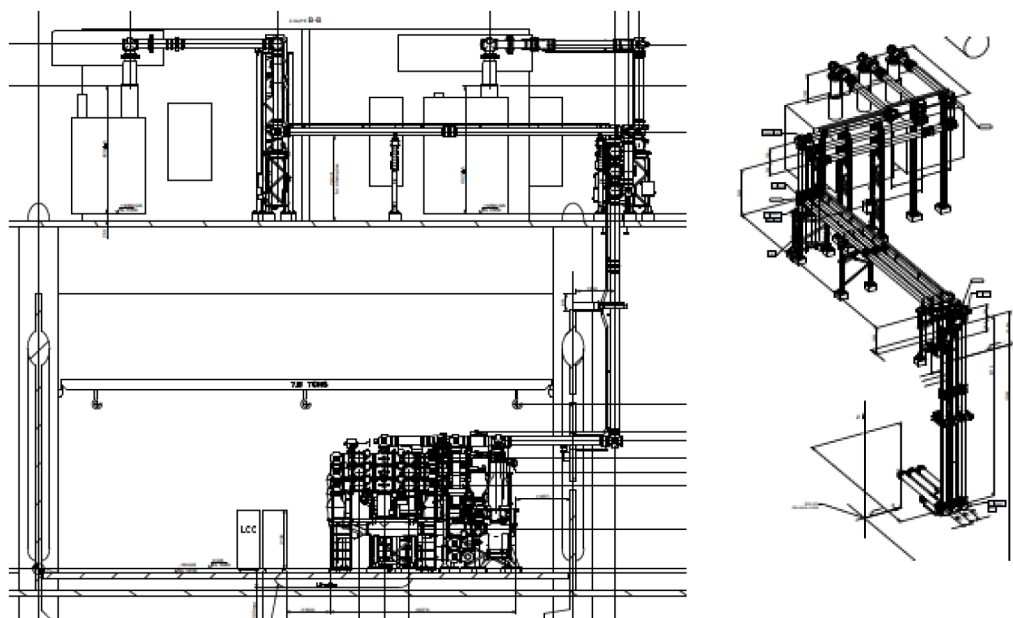


Figure 4—Conventional Design with Gas Insulated Busducts (GIBs)

Advantages of EHV Cables over Conventional GIBs achieved on the project

- Utilizing 230kV cables in lieu of the Gas Insulated Bus Ducts (GIB), which uses SF₆ gas. SF₆ is a potent GHG that is widely used as dielectric material for high electric voltage. Replacing GIB with 230 KV cables was expected to reduce the major portion of GIB and hence the amount of SF₆

gas requirement resulting in effectively reducing SF6 gas requirement to de-risk project schedule and reduce OPEX and CAPEX.

- Reduction / Elimination of the CAPEX for Company, associated with the multiple maintenance visits for the GIB joints post energization of the facility.
- Reduction / Elimination of the OPEX for Company, associated with the multiple visits required post energization for clearing the SF6 GAS alarms coming out of the outdoor GIBs.
- Cable Solution provides overall better Planned Preventive Maintenance (PPM) for the GIS system.
- Cables provide flexibility of making adjustments during final connection with equipment. GIBs have low tolerance for mismatches at site.
- There is no limitation of deflection in structural supports for cables and there no need to dismantle the connections during load-out and transportation.
- Cables offer quicker installation and commissioning sequences at site.
- Risk due to the usual long lead delivery of the GIBs (that are manufactured by GIS OEM sub-vendors) can be avoided, which will support creating a float under fabrication.
- The risk of the missing accessories for the GIBs at yard from the vendor during the erection time will be eliminated.
- Elimination of the risk of multiple issues faced during the tightening of the GIB flange joints which results in possible re-work during erection.
- Reduction in the requirement of Company power operation representatives for the witness of GIB gas leak testing at Contractor yard.
- Reduction / Elimination of the platform shutdowns due to tripping of the GIS due to SF6 leak cases as per the feedback from Company site team.

Comparison of Lengths for GIB vs Cables

Table 1—Comparison of GIB vs Cable Option

Sl. No.	GIB Lengths (per phase – approx.)	Platform-5			Platforms-1,3,4,6		
		Base Case - All GIB (m)	Cable Solution (m)	% of GIB saving per phase w.r.t. base option	Base Case - All GIB (m)	Cable Solution (m)	% of GIB saving per phase w.r.t. base option
				Selected Option			Selected Option
1	GIS to Transformer-1	55	3	94.55	35	5	85.71
2	GIS to Transformer-2	55	3	94.55	41	5	87.80
3	GIS to Shunt Reactor-1	40	3	92.50	-	-	N/A
4	GIS to Shunt Reactor-2	40	3	92.50	-	-	N/A

Statement of Theory and Definitions (Design Aspects)

Feasibility of replacing 230kV Gas Insulated Busduct design with Cables requires holistic review to identify major impacts on the offshore platform and associated equipment. The following design aspects must be reviewed in detail to avoid issues during installation, testing and commissioning:

Cable Size

Shunt reactors were rated as 120MVAR, while Transformers are 60/80MVA. Therefore, cables rated for 120MVA (i.e. 301A) were found to be adequate for both equipment. Accordingly, 3x1Cx630 sq.mm. Cable with an ampacity of more than 600A and short circuit rating of 63kA (1 sec) was selected as per utility standard for 230kV Cables (refer fig.6).

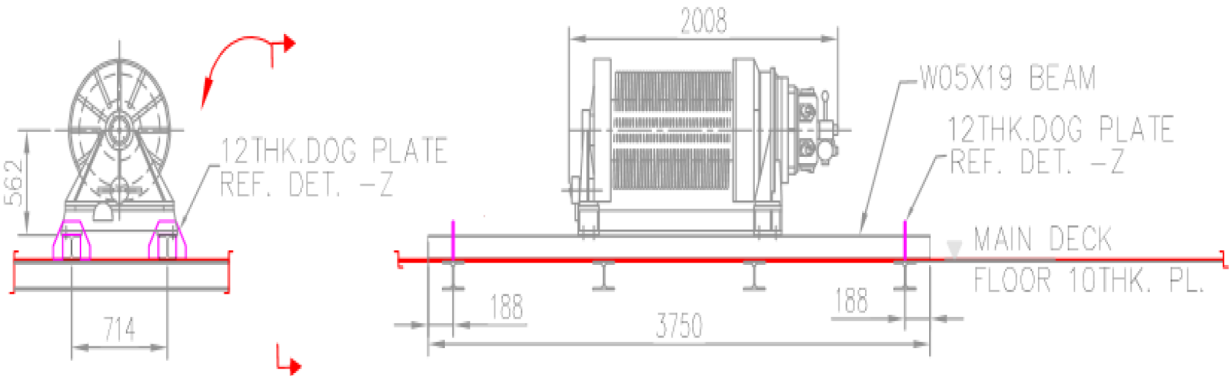


Figure 5—Tugger Assembly

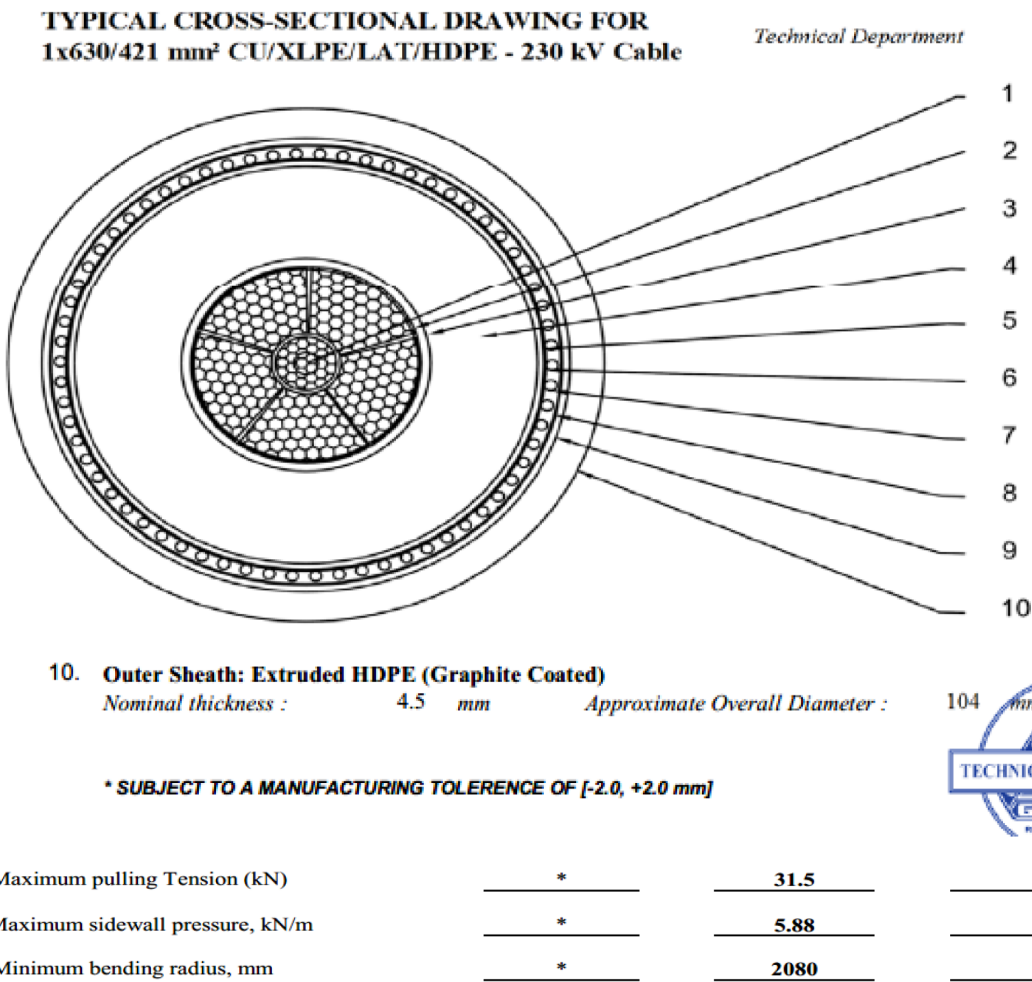


Figure 6—230kV Topside Cable Details

Minimum Bending Radius

230kV Cable Minimum Bending Radius had to be considered in layout to avoid any damage to the cable. Additional space is required due to the requirement of MBR at bending locations. This requirement had to be coordinated with the cable vendor.

Note: Clearance required shall be calculated based on the Minimum Bending Radius specified by 230kV Cable Supplier. Accordingly, spacing on the Cable Supports shall be decided based on 230kV Cable Supplier data.

Cable Termination Kit

Termination of cable at transformer, shunt reactor and GIS ends require space for preparing and installing termination kits. Also, scope for male & female components of the termination had to be reviewed between Transformer/Shunt Reactor, GIS and Cable vendors, since this is a specialized activity performed by certified jointers.

Cable supports

230kV Cables as per utility standards are unarmored and require support. Cable support / ladders shall also comply with the requirements of MBR. The cable supports shall be in conformance with the severe corrosive condition especially when installed outdoor and shall withstand forces while pulling.

Cable Clamps

Single core cables need to be laid in trefoil or flat formation and require clamps and fixing arrangement. Cable clamps shall be suitable for the severe corrosives condition. However, flat formation of laying was considered due to space constraints at offshore.

Power Transformer / Shunt Reactor Interface

Termination of cables at transformer and shunt reactor ends requires enclosures for arrangement of oil-to-oil interface. This results in an increase in overall equipment dimension. Also, entry of cable needs to be from side or bottom as top entry will not be feasible as per vendors and will require excessive height for supporting arrangement. This would result in major design change since GIB interface with this equipment is traditionally from the top. This interface had to be coordinated with Power Transformer / Shunt Reactor vendors.

GIS interface

Cable entry through switchgear room wall or floor also impacted the GIS interface. This had to be reviewed and coordinated with GIS vendor. Wall/floor penetrations and sealing arrangements (fire stop) had to be designed, supplied & certified by Building Subcontractor.

Cable Installation Feasibility

Pulling and installing subsea cables at topsides at offshore platforms maintaining Minimum Bending Radius is a major challenge. Space is required for the installation of temporary installation equipment such as winches, tuggers, quadrants, rollers etc.

Description and Application of Equipment and Processes

Detailed analysis requires various inputs for various items involved in the EHV Power Distribution system. Specific checkpoints included the following:

Cable Data

Cable Overall Diameter: 104mm, Minimum Bending Radius: 2080mm, Refer [Fig 6](#) below for 230kV Cable Data.

Transformer Data

Transformer Outline Drawing: Location of HV Bushing (Oil-Oil interface). Refer [Fig.7](#) below.

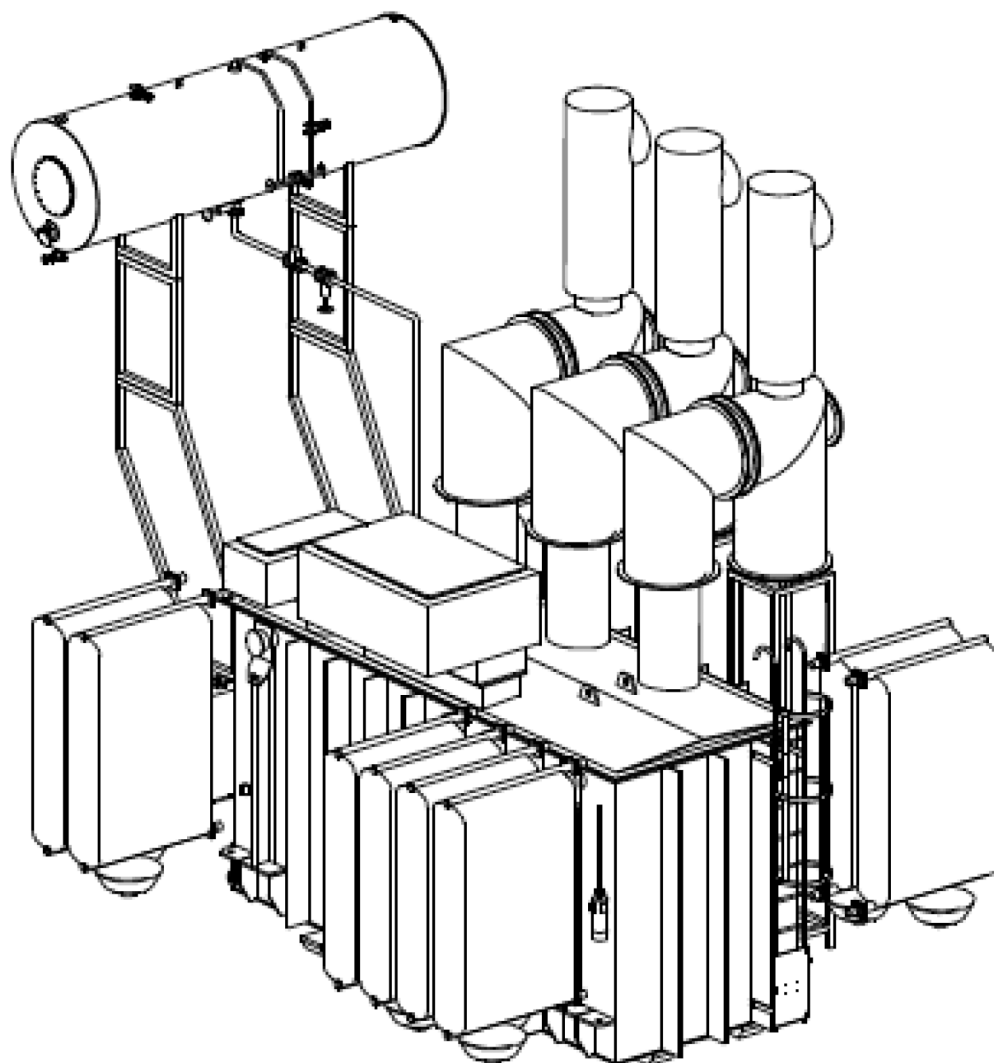


Figure 7—Power Transformer with 230kV Cable Box

Shunt Reactor Data

Reactor Outline Drawing: Location of HV Bushing (Oil-Oil interface). Refer [Fig.8](#) below.

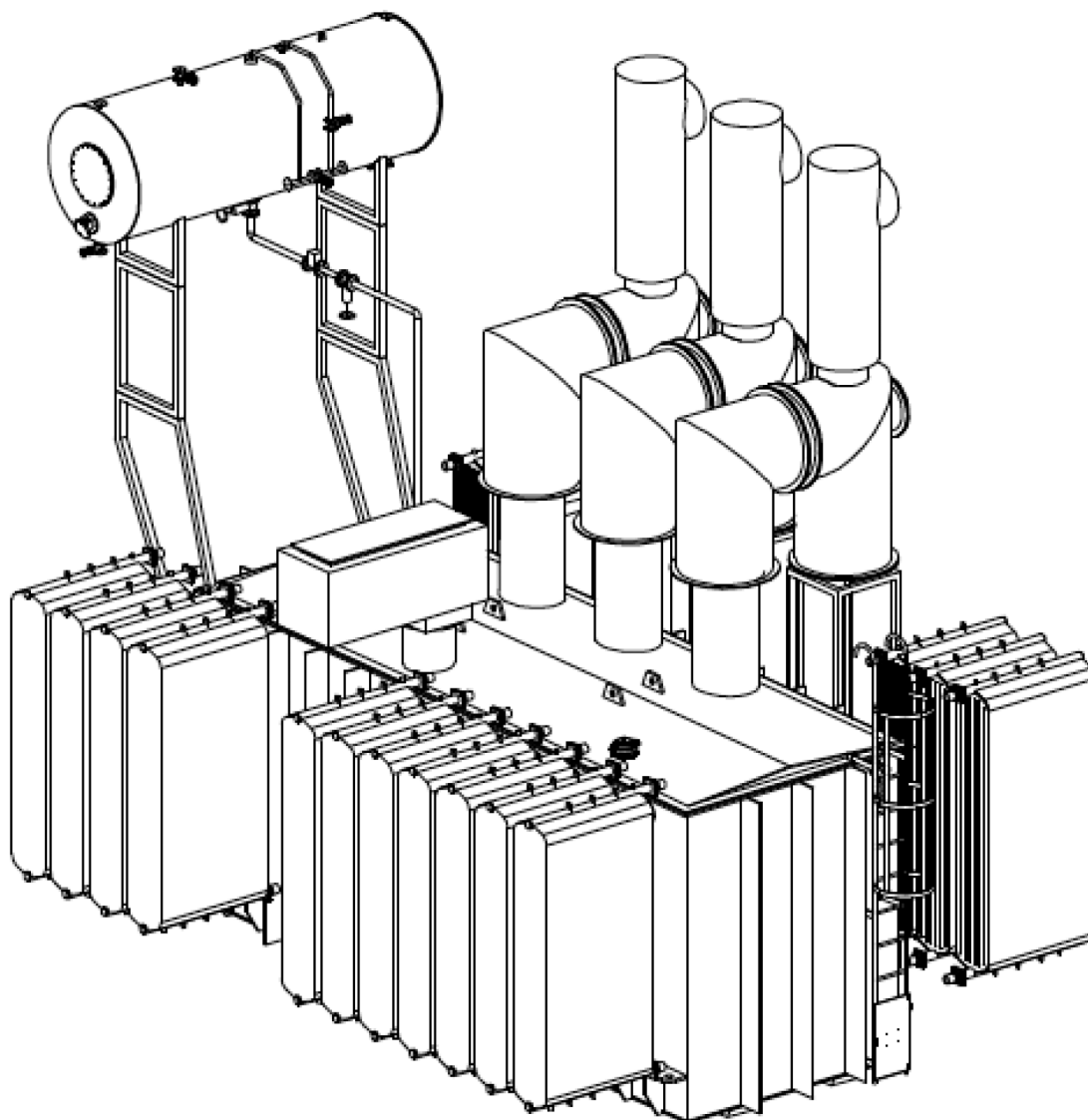


Figure 8—Shunt Reactor with 230kV Cable Box

GIS Data

GIS Outline Drawing: Required to verify the Cable interface

Also, the space for Cable Termination inside the Switchgear Room needs to be carefully verified.

Switchgear Room and Transformer/Shunt Reactors

Feasibility of top/side entry at GIS to be verified

Feasibility of bottom entry at GIS to be verified

Feasibility of bottom entry at Transformer & Shunt Reactor to be verified

Feasibility of side entry at Transformer & Shunt Reactor to be verified

Cable Pulling Strategy

Proper spacing of tuggers and quadrants is critical for safe cable pulling. This requires close coordination with offshore teams. Refer a typical scheme (Fig-9) which depicts the location of all the pulling equipment.

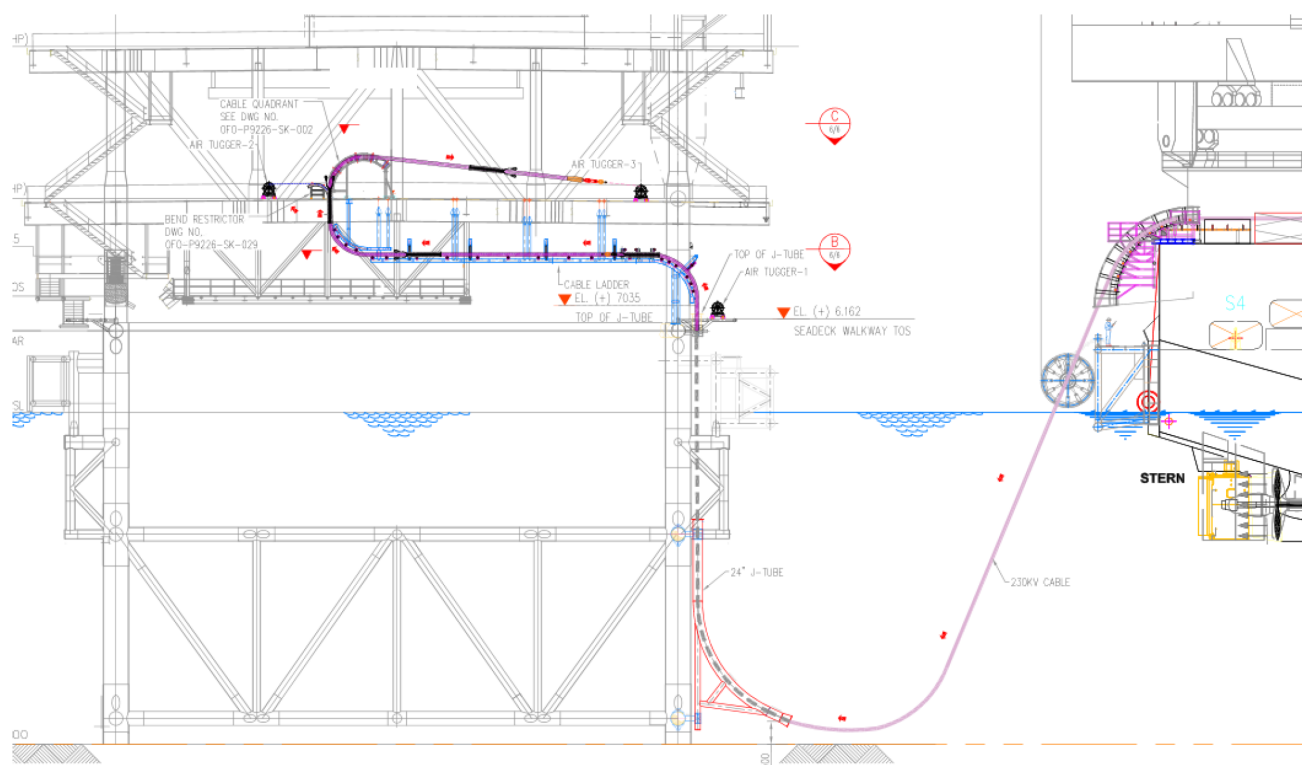


Figure 9—Cable Pulling Scheme using Tuggers & Quadrants

Presentation of Data and Results

During the detailed engineering stage all aspects of submarine cable design were studied in detail. A methodical approach was adopted to cover each design review step carefully. All stakeholders (both internal and external) were consulted. Overall cost and project schedule were adequately considered while making decisions. Now, let us review each design review stage.

Submarine Cables Interface with GIS at Platform-5

To avoid the offshore work and requirement of cable joint, submarine cables were terminated at GIS cable seal compartment below the switchgear room. Hybrid solution with short (approx. 6-7m) GIB will be required (Refer Fig-10 below).

Note: GIB could have been avoided if there was a false floor inside GIS room.

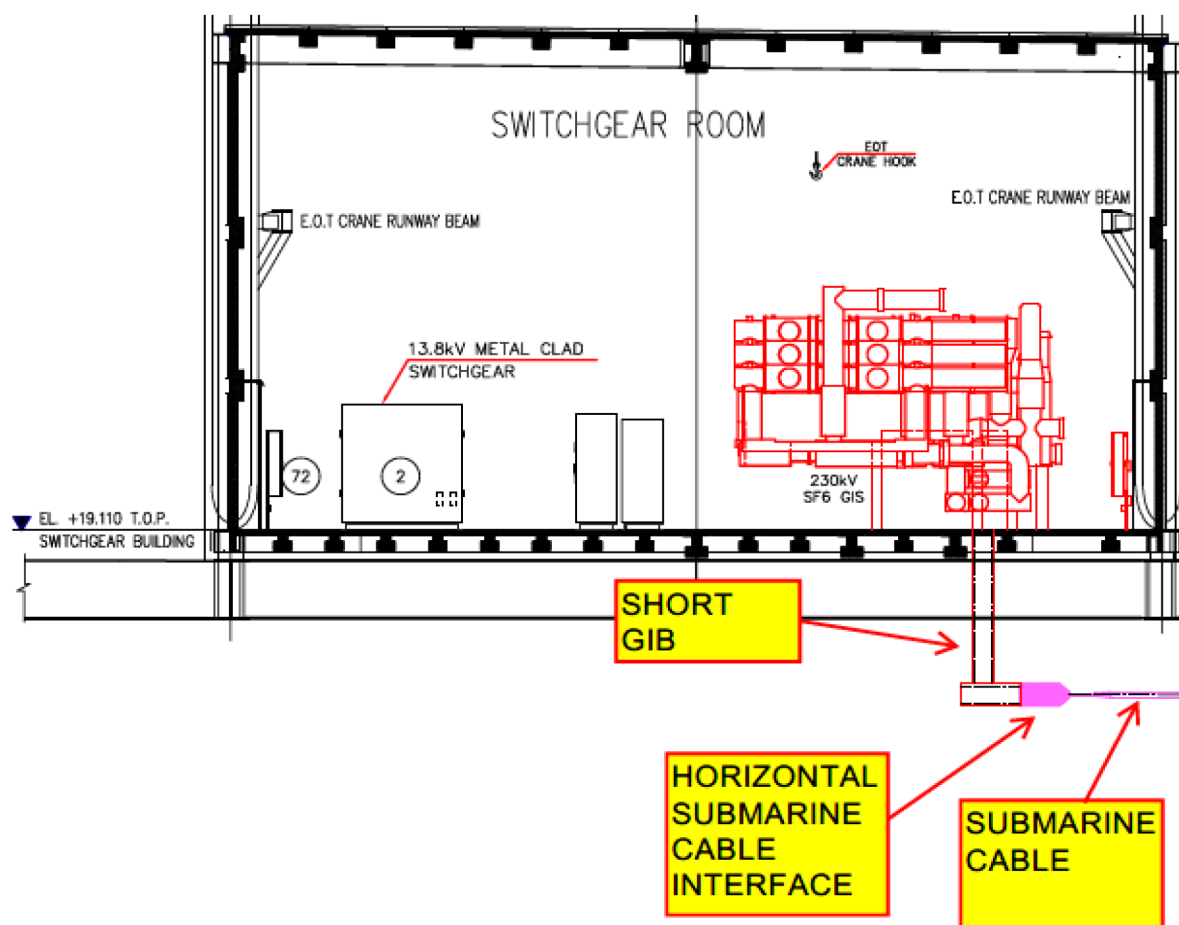


Figure 10—Hybrid solution with short L Shaped GIB

Transformer/Shunt Reactor interface with GIS (top entry)

Transformer/Shunt Reactors can have only bottom entry as per vendors. Also, top interface with GIB cable Box is not feasible as per vendor. Additional space will be required below upper deck for feasibility of cable support installation and cable pulling. Hybrid Solution with GIS Cable Box side entry (outside the switchgear room) and Transformer/Shunt Reactors bottom Entry. Short GIB, Cable Termination and Cable Bending will require long supporting arrangement (Refer Fig 11 below).



Side entry will require a cable joint and involve topside routing which will have space constraint. Therefore, hybrid solution with short GIB section (approx. 6-7m) had to be used to avoid the offshore work and requirement of cable joint, submarine cables will be terminated at GIS cable seal compartment below the switchgear room (Refer [Fig.12](#) below).



Transformer interface with GIS (bottom entry – plan)

Top or side entry was not feasible as per vendors. Based on the orientation of transformers, side entry with cable solution required cables of one transformer to pass through area of other transformer which could pose redundancy concern in case of fire incidence on one transformer. Therefore, bottom entry at transformer had to be studied further. Bottom entry at both GIS and Transformer ends with hybrid solution at GIS end was the most feasible solution (Refer Fig.13 below).

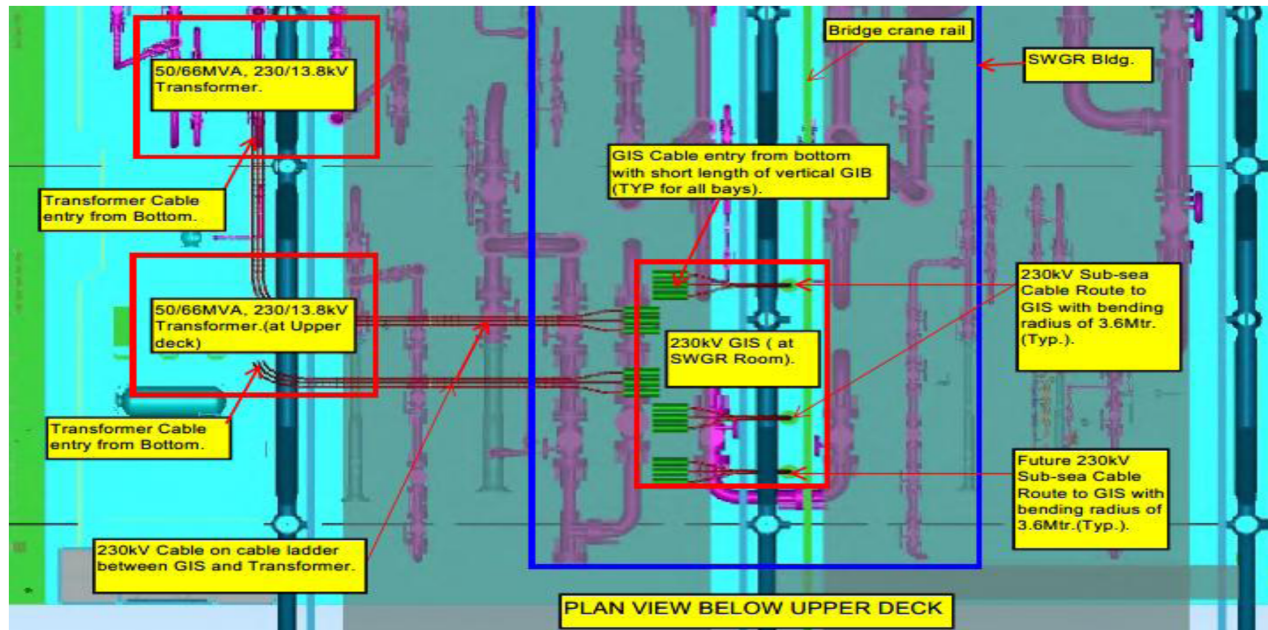


Figure 13—Hybrid solution with bottom entry at Power Transformers

Observation Summary

Following is the list of observations based on careful review of each point:

- Refer Fig. 7 & 8 for feasibility of bottom entry with HV Bushing (Oil-to-Oil interface).
- Additional space was required below the Transformers & Shunt deck, for feasibility of cable support installation and cable pulling.
- To avoid the offshore work and requirement of cable joint, submarine cables must be terminated at GIS cable seal compartment below the switchgear room. This was the hybrid solution with short GIB (approx. 6-7m), Refer Fig. 10 & 12 or respective platforms.
- Side entry at Transformers and Shunt Reactors is not feasible due to space constraint. There will be a clash with firewall, interference with escape route and feasibility issues at shunt reactor end due to limited space.
- Bottom entry at GIS end as well as transformer end was most feasible option for platform-1/2/3/4 (Refer Fig. 12 & 13)
- Short GIB for transformer feeders could be avoided if cables can be directly terminated at GIS inside switchgear room, if there was enough space to prepare for the cable termination. This was implemented on the next project executed by the team.

Conclusions

EHV power system design requires a comprehensive detailed study. A methodical approach should be adopted to address each design review point and to ensure that all design limitations for use of EHV cables

are duly addressed. The following is the list of conclusions after the detailed study and implementation of EHC cables instead of GIBs:

- Replacing GIB with 230 KV cables resulted in a reduction in the length of GIB portion by 2/3rd and hence the amount of SF6 gas requirement resulting in effectively reducing SF6 gas requirement to de-risk project schedule (by approx. 6months) and reduce OPEX and CAPEX (approx. one million USD). CO2 Avoidance was estimated at approx. 47,674 tCO2e.
- Minimum deck to deck height shall be 12m to allow 2 nos. 90-degree bends of the subsea cables. Otherwise, design needs to be reviewed carefully to explore other options such as breaking out and routing only the power cores with less MBR.
- Hybrid solution (short GIB portion) for GIS interface was considered, as that was found to be the most feasible option based on available space on the platform. This can also be avoided if there is a false/raised floor within the switchgear room.
- 230kV Cable were successfully used in place of GIBs for Transformer and Shunt Reactor Feeders.
- Upper deck at Platform-5 where transformers and shunt reactors were located, the primary framing beam depth had to be suitably reduced at the localized portion for the feasibility of cable installation and termination.
- Transformer feeders were side entry at GIS side for platform no. 2. This option required major structural supports since Power Transformers and Shunt Reactors were located on the deck above.
- For Platform-1/3/4/6, bottom entry was used for Submarine Cable feeders as well as Transformer feeders.
- Plug-in dry type termination kits (oil to oil type) were used for ease of installation, operation & maintenance.
- Unarmored Cables were utilized for ease of bending and to lower MBR as per utility standards
- Cables were in flat formation instead of trefoil formation due to space constraint and ease of installation.
- Cables were laid on cable ladders except at bending location where structural support was considered.
- HV testing of 230kV cables was carried out from GIS side.

Acknowledgements

Design of EHV Power Distribution System is a complex activity that requires close coordination with other stakeholders such as Offshore Team, Client and Vendors. This paper would not have been possible without the support of the following stakeholders.

- NMDC ENERGY Engineering Management for providing continuous support and resources to manage mega packages.
- Disciplines / Departments – Electrical, Pipeline, Piping, Structural, Offshore, Digital Twin Team & PMT for inputs to Submarine Cable Length Estimation and Geotechnical Investigation.
- Client – Project Management Team for participating in review and approval stages.
- Vendors/Sub-Contractors – Submarine Cable, Transformers, Reactors, Building Sub-Contractors.
- Power System Study Contractor – Subcontractor for Electrical Studies.

Abbreviations

ADIPEC Abu Dhabi International Petroleum Exhibition & Conference
AI Artificial Intelligence

CAPEX	Capital Expenditure
CMMS	Computerized Maintenance Management System
CO ₂	Carbon Dioxide
DCS	Distributed Control System
EHC	Extra High Current (verify if intended; might be a typo for EHV)
EHV	Extra High Voltage
EN	European Norm
EPC	Engineering, Procurement, and Construction
EPR	Ethylene Propylene Rubber
ESPs	Electric Submersible Pumps
FEED	Front End Engineering Design
FO	Fiber Optic
GHG	Greenhouse Gas
GIB	Gas Insulated Busduct
GIS	Gas Insulated Switchgear
GWP	Global Warming Potential
HV	High Voltage
HVAC	Heating, Ventilation and Air Conditioning
IEC	International Electrotechnical Commission
IoT	Internet of Things
kV	Kilovolt
kVA	Kilovolt-Ampere
M	Meter
MVA	Megavolt-Ampere
MVAR	Megavolt-Ampere Reactive
MBR	Minimum Bending Radius
ML	Machine Learning
NMDC	National Marine Dredging Company
OEM	Original Equipment Manufacturer
OPEX	Operational Expenditure
PMT	Project Management Team
PPM	Planned Preventive Maintenance
RPA	Robotic Process Automation
SCADA	Supervisory Control and Data Acquisition
SF ₆	Sulfur Hexafluoride
SPEL	Smart Plant Electrical
tCO ₂ e	Tonnes of Carbon Dioxide Equivalent
U.A.E.	United Arab Emirates
VSD	Variable Speed Drive
WHT	Well Head Tower
XLPE	Cross-Linked Polyethylene